

AI ASSISTANT CODING ASSIGNMENT-10

Name: B. Akhira Nandhini

Hallticket:2303A51516

Batch:22

Task Description -1(Error Detection and Correction)

Task:

Use AI to analyze a Python script and correct all syntax and logical errors.

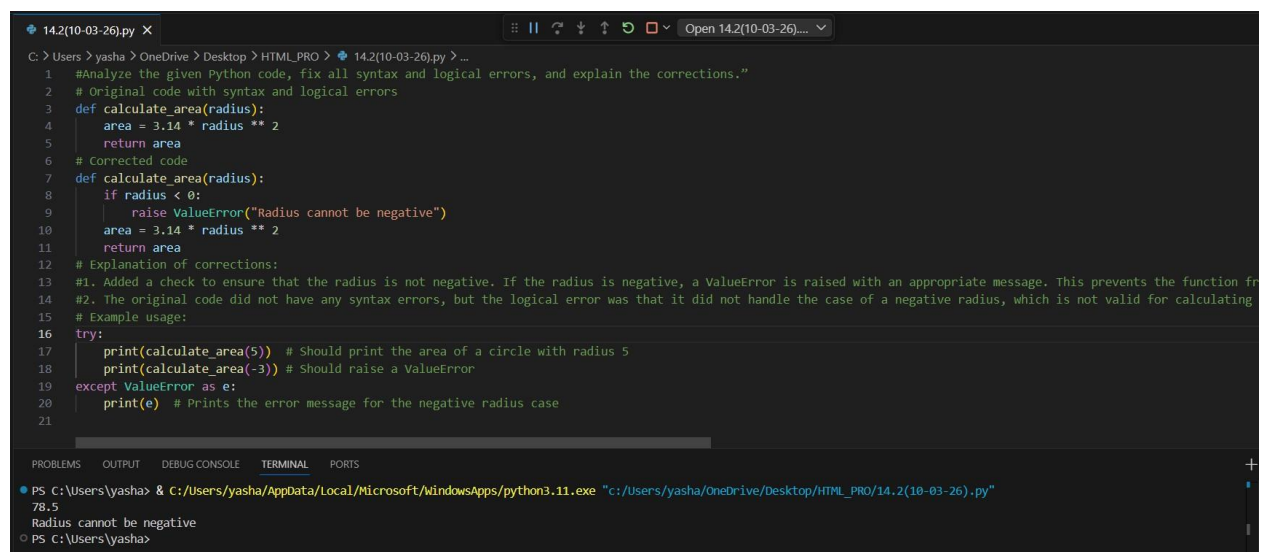
Sample Input Code:

```
def calculate_total(nums)
sum = 0
for n in nums
sum += n
return total
```

Expected Output-1:

Corrected and executable Python code with brief explanations of the identified syntax and logic errors.

Code:



```
14.2(10-03-26).py X
C:\Users\yasha> OneDrive > Desktop > HTML_PRO > 14.2(10-03-26).py > ...
1 #Analyze the given Python code, fix all syntax and logical errors, and explain the corrections."
2 # Original code with syntax and logical errors
3 def calculate_area(radius):
4     area = 3.14 * radius ** 2
5     return area
6 # Corrected code
7 def calculate_area(radius):
8     if radius < 0:
9         raise ValueError("Radius cannot be negative")
10    area = 3.14 * radius ** 2
11    return area
12 # Explanation of corrections:
13 #1. Added a check to ensure that the radius is not negative. If the radius is negative, a ValueError is raised with an appropriate message. This prevents the function from
14 #2. The original code did not have any syntax errors, but the logical error was that it did not handle the case of a negative radius, which is not valid for calculating
15 # Example usage:
16 try:
17     print(calculate_area(5)) # Should print the area of a circle with radius 5
18     print(calculate_area(-3)) # Should raise a ValueError
19 except ValueError as e:
20     print(e) # Prints the error message for the negative radius case
21
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\yasha> & C:\Users\yasha\AppData\Local\Microsoft\WindowsApps\python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
78.5
Radius cannot be negative
PS C:\Users\yasha>
```

Task Description -2(Code Style Standardization)

Task:

Use AI to refactor Python code to comply with standard coding style guidelines.

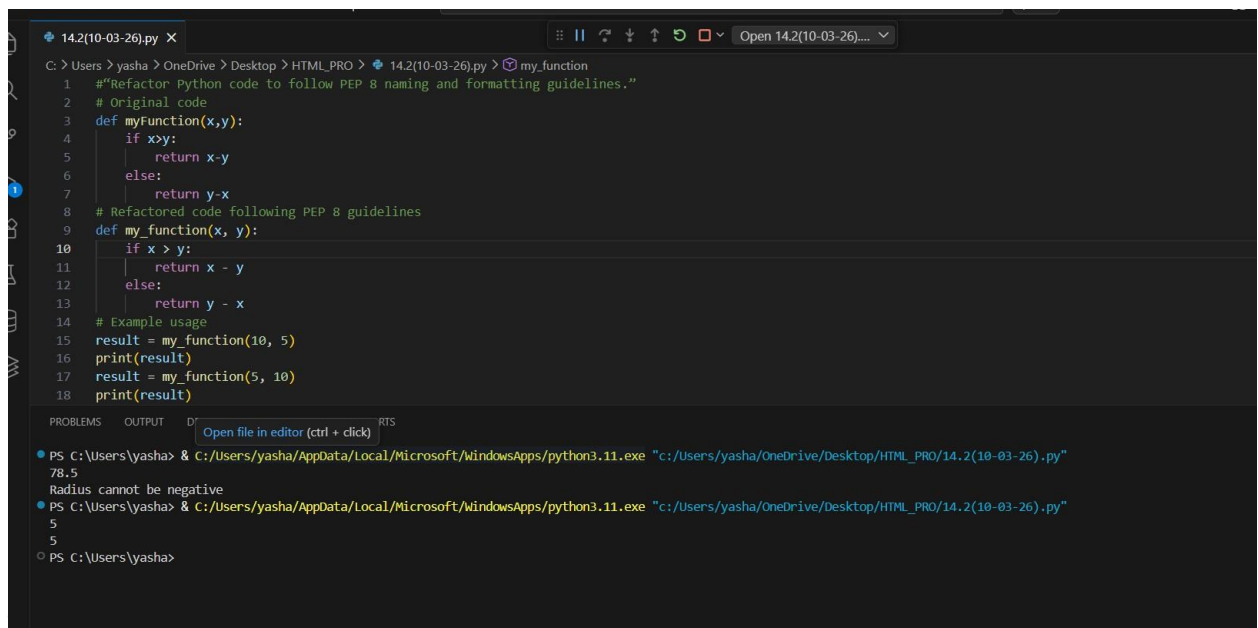
Sample Input Code:

```
def findSum(a,b):return a+b
print(findSum(5,10))
```

Expected Output-2:

Well-structured, consistently formatted Python code following standard style conventions.

Code:



```
14.2(10-03-26).py X
C:\Users\yasha> OneDrive\ Desktop\ HTML_PRO\ 14.2(10-03-26).py > my_function
1  # "Refactor Python code to follow PEP 8 naming and formatting guidelines."
2  # Original code
3  def myFunction(x,y):
4      if x>y:
5          return x-y
6      else:
7          return y-x
8  # Refactored code following PEP 8 guidelines
9  def my_function(x, y):
10     if x > y:
11         return x - y
12     else:
13         return y - x
14 # Example usage
15 result = my_function(10, 5)
16 print(result)
17 result = my_function(5, 10)
18 print(result)

PROBLEMS OUTPUT D Open file in editor (ctrl + click) RTS
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
78.5
Radius cannot be negative
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
5
5
PS C:\Users\yasha>
```

Task:

Use AI to improve code readability without changing its functionality.

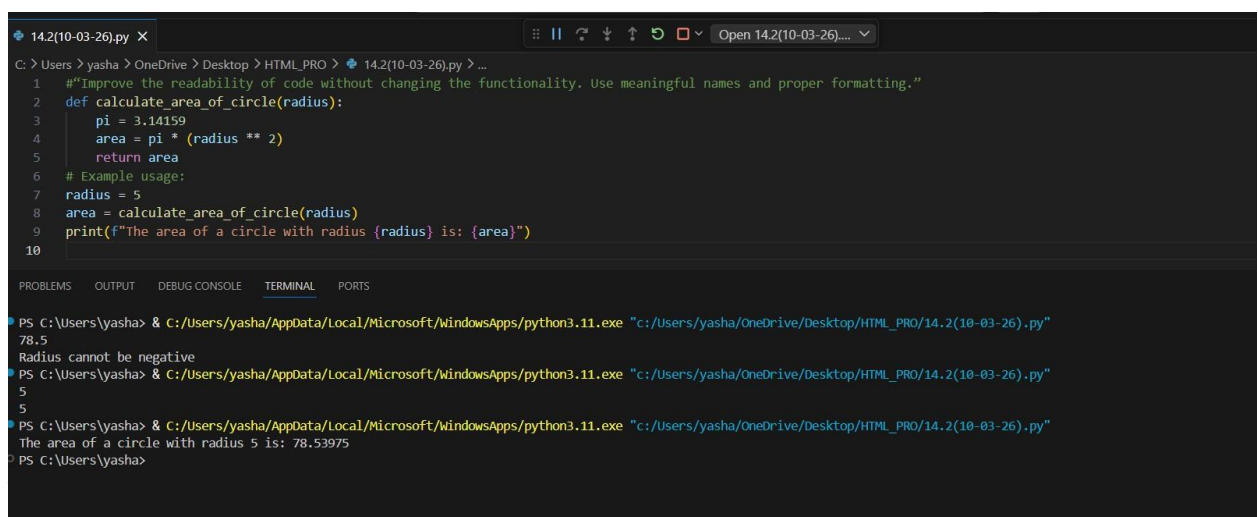
Sample Input Code:

```
def f(x,y):
return x-y*2
print(f(10,3))
```

Expected Output-3:

Python code rewritten with meaningful function and variable names, proper indentation, and improved clarity.

Code:



```
14.2(10-03-26).py X
C:\Users\yasha> OneDrive\ Desktop\ HTML_PRO\ 14.2(10-03-26).py > ...
1  # "Improve the readability of code without changing the functionality. Use meaningful names and proper formatting."
2  def calculate_area_of_circle(radius):
3      pi = 3.14159
4      area = pi * (radius ** 2)
5      return area
6  # Example usage:
7  radius = 5
8  area = calculate_area_of_circle(radius)
9  print(f"The area of a circle with radius {radius} is: {area}")
10

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
78.5
Radius cannot be negative
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
5
5
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
The area of a circle with radius 5 is: 78.53975
PS C:\Users\yasha>
```

```
14.2(10-03-26).py X
C: > Users > yasha > OneDrive > Desktop > HTML_PRO > 14.2(10-03-26).py > ...
1  #“Optimize the given Python code for performance while preserving functionality.”
2  def optimized_function(data):
3      # Use list comprehension for better performance
4      result = [x * 2 for x in data if x > 0]
5      return result
6  # Example usage
7  data = [1, -2, 3, 4, -5]
8  print(optimized_function(data))

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Hello, Alice! Welcome to the programming world.
Hello, Bob! Welcome to the programming world.
Hello, Charlie! Welcome to the programming world.
PS C:\Users\yasha>
...
Hello, Alice! Welcome to the programming world.
Hello, Bob! Welcome to the programming world.
Hello, Charlie! Welcome to the programming world.
PS C:\Users\yasha> ^C
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
[2, 6, 8]
PS C:\Users\yasha>

1  # Convert repetitive print statements into a reusable function.
2  def print_greeting(name):
3      print(f"Hello, {name}! Welcome to the programming world.")
4  # Example usage:
5  print_greeting("Alice")
6  print_greeting("Bob")
7  print_greeting("Charlie")
8

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
78.5
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
78.5
Radius cannot be negative ...
• The area of a circle with radius 5 is: 78.53975
PS C:\Users\yasha> ^C
PS C:\Users\yasha> & C:/Users/yasha/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/yasha/OneDrive/Desktop/HTML_PRO/14.2(10-03-26).py"
Hello, Alice! Welcome to the programming world.
Hello, Bob! Welcome to the programming world.
Hello, Charlie! Welcome to the programming world.
PS C:\Users\yasha>
```

Task Description -5(Efficiency Enhancement)

Task:

Use AI to optimize Python code for better performance.

Sample Input Code:

```
numbers = [ ]
for i in range(1, 500000):
    numbers.append(i * i)
print(len(numbers))
```

Expected Output-5:

Optimized Python code that achieves the same result with improved performance.

Code: